COMPUTER ENGINEERING
QUEZON CITY

COURSE CODE/TITLE:	CPE212 - Automating Server Management	**SECTION:**	CPE31S2
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1.1 HANDS-ON ACTIVITY — CREATING VIRTUAL MACHINES

1. PROCEDURE OUTPUT

Task 1: Do the following on Server 1, Server 2, and Local Machine. In editing the file using nano command, press control + O to write out (save the file). Press enter when asked for the name of the file. Press control + X to end.

1. Change the hostname using the command sudo nano /etc/hostname

1.1 Use server1 for Server 1

```
  GNU nano 7.2                          /etc/hostname *
server1
```

1.2 Use server2 for Server 2

```
  GNU nano 7.2                          /etc/hostname *
server2
```

1.3 Use workstation for the Local Machine

```
  GNU nano 7.2                          /etc/hostname *
workstation
```

2. Edit the hosts using the command sudo nano /etc/hosts. Edit the second line.

2.1 Type 127.0.0.1 server 1 for Server 1

```
  GNU nano 7.2                          /etc/hosts *
127.0.0.1 localhost
127.0.0.1 server1

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
ff02::3 ip6-allhosts
```

2.2 Type 127.0.0.1 server 2 for Server 2

```
  GNU nano 7.2                                /etc/hosts *
127.0.0.1 localhost
127.0.0.1 server2

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
ff02::3 ip6-allhosts
```

2.3 Type 127.0.0.1 workstation for the Local Machine

```
  GNU nano 7.2                                /etc/hosts *
127.0.0.1 localhost
127.0.0.1 workstation

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
ff02::3 ip6-allhosts
```

Task 2: Configure SSH on Server 1, Server 2, and Local Machine. Do the following:
1. Upgrade the packages by issuing the command sudo apt update and sudo apt upgrade respectively.

```
queleroyubuntu3@QueLeroyubuntu3:~/Desktop$ sudo apt update
Get:1 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Hit:2 http://ph.archive.ubuntu.com/ubuntu jammy InRelease
Get:3 http://ph.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
```

```
queleroyubuntu3@QueLeroyubuntu3:~/Desktop$ sudo apt update
Get:1 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Hit:2 http://ph.archive.ubuntu.com/ubuntu jammy InRelease
Get:3 http://ph.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
```

```
queleroyubuntu3@QueLeroyubuntu3:~/Desktop$ sudo apt update
Get:1 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Hit:2 http://ph.archive.ubuntu.com/ubuntu jammy InRelease
Get:3 http://ph.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
```

```
queleroyubuntu3@QueLeroyubuntu3:~/Desktop$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

```
queleroyubuntu3@QueLeroyubuntu3:~/Desktop$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

```
queleroyubuntu3@QueLeroyubuntu3:~/Desktop$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

2. Install the SSH server using the command sudo apt install openssh-server.

```
queleroyubuntu3@QueLeroyubuntu3:~$ sudo apt install openssh-server
[sudo] password for queleroyubuntu3:
Reading package lists... Done
Building dependency tree... Done
```

```
queleroyubuntu3@QueLeroyubuntu3:~$ sudo apt install openssh-server
[sudo] password for queleroyubuntu3:
Reading package lists... Done
Building dependency tree... Done
```

```
queleroyubuntu3@QueLeroyubuntu3:~$ sudo apt install openssh-server
[sudo] password for queleroyubuntu3:
Reading package lists... Done
Building dependency tree... Done
```

3. Verify if the SSH service has started by issuing the following commands:

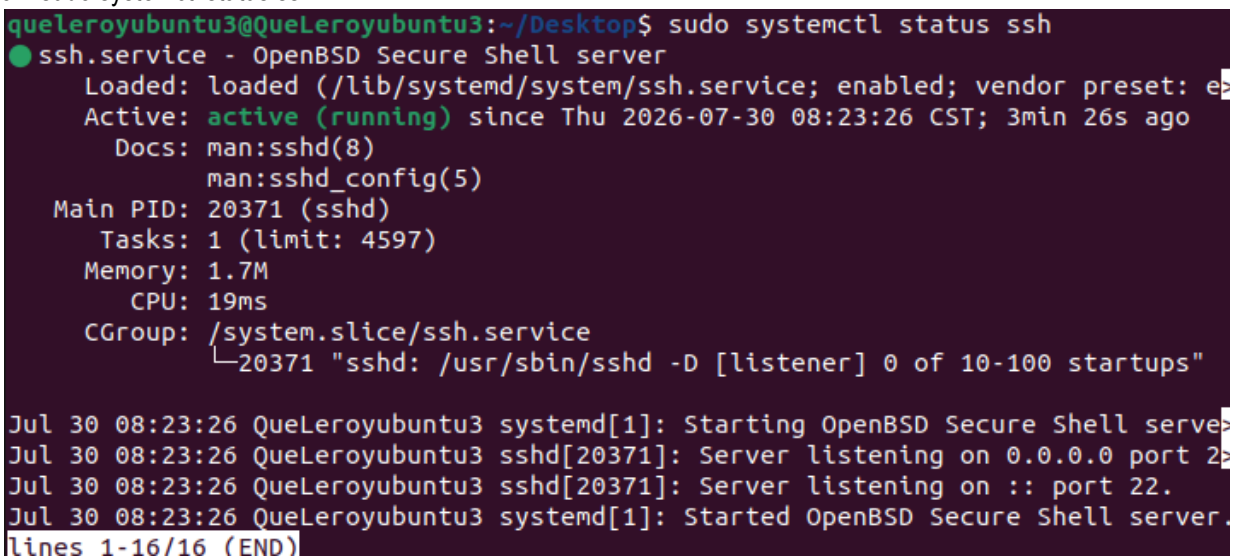
3.1 sudo service ssh start

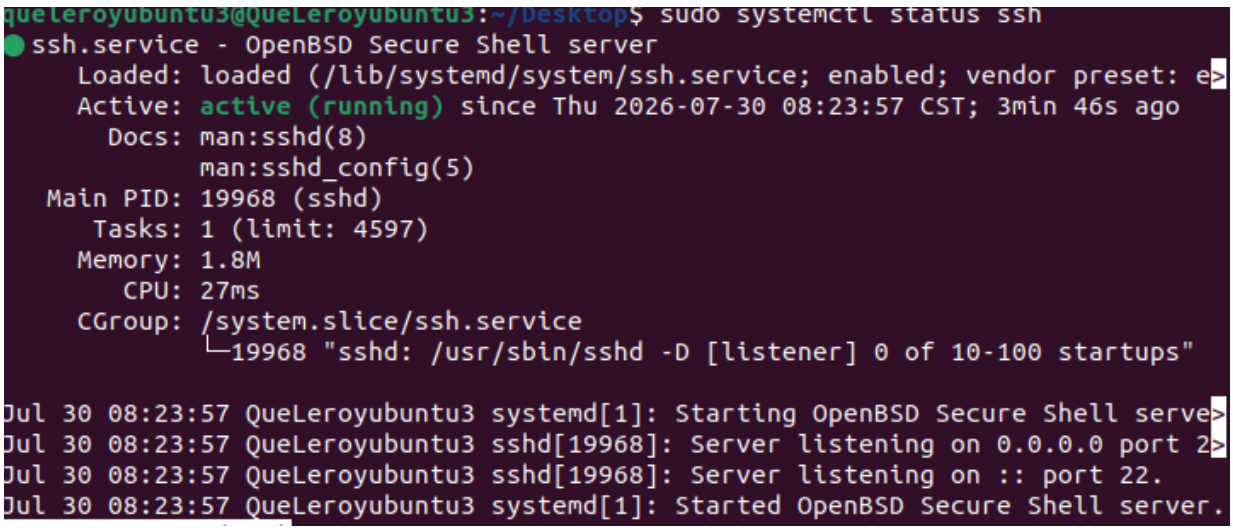
```
queleroyubuntu3@QueLeroyubuntu3:~/Desktop$ sudo service ssh start
```

```
queleroyubuntu3@QueLeroyubuntu3:~/Desktop$ sudo service ssh start
```

```
queleroyubuntu3@QueLeroyubuntu3:~/Desktop$ sudo service ssh start
```

3.2 sudo systemctl status ssh





```
queleroyubuntu3@QueLeroyubuntu3:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: e>
     Active: active (running) since Thu 2026-07-30 08:24:14 CST; 3min 59s ago
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 20514 (sshd)
      Tasks: 1 (limit: 4597)
     Memory: 1.7M
        CPU: 21ms
     CGroup: /system.slice/ssh.service
             └─20514 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 30 08:24:14 QueLeroyubuntu3 systemd[1]: Starting OpenBSD Secure Shell serve>
Jul 30 08:24:14 QueLeroyubuntu3 sshd[20514]: Server listening on 0.0.0.0 port 2>
Jul 30 08:24:14 QueLeroyubuntu3 sshd[20514]: Server listening on :: port 22.
Jul 30 08:24:14 QueLeroyubuntu3 systemd[1]: Started OpenBSD Secure Shell server.
```

4. Configure the firewall to all port 22 by issuing the following commands:

4.1 sudo ufw allow ssh

```
queleroyubuntu3@QueLeroyubuntu3:~/Desktop$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
```

```
queleroyubuntu3@QueLeroyubuntu3:~/Desktop$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
```

```
queleroyubuntu3@QueLeroyubuntu3:~/Desktop$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
```

4.2 sudo ufw enable

```
queleroyubuntu3@QueLeroyubuntu3:~/Desktop$ sudo ufw enable
Firewall is active and enabled on system startup
```

```
queleroyubuntu3@QueLeroyubuntu3:~/Desktop$ sudo ufw enable
Firewall is active and enabled on system startup
```

```
queleroyubuntu3@QueLeroyubuntu3:~/Desktop$ sudo ufw enable
Firewall is active and enabled on system startup
```

4.3 sudo ufw status

```
queleroyubuntu3@QueLeroyubuntu3:~/Desktop$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)
```

```
queleroyubuntu3@QueLeroyubuntu3:~/Desktop$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)
```

```
queleroyubuntu3@QueLeroyubuntu3:~/Desktop$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)
```

Task 3: Verify network settings on Server 1, Server 2, and Local Machine. On each device, do the following:

1. Record the ip address of Server 1, Server 2, and Local Machine. Issue the command ifconfig and check network settings. Note that the ip addresses of all the machines are in this network 192.168.56.XX.

1.1 Server 1 IP address: 192.168.56.9

```
inet 192.168.56.9
```

1.2 Server 2 IP address: 192.168.56.10

```
inet 192.168.56.10
```

1.3 Server 3 IP address: 192.168.56.11

```
inet 192.168.56.11
```

2. Make sure that they can ping each other.

2.1 Connectivity test for Local Machine 1 to Server 1: ✅ Successful ☐ Not Successful

```
queleroyubuntu3@workstation:~/Desktop$ ping 192.168.56.9
PING 192.168.56.9 (192.168.56.9) 56(84) bytes of data.
64 bytes from 192.168.56.9: icmp_seq=1 ttl=64 time=2.04 ms
64 bytes from 192.168.56.9: icmp_seq=2 ttl=64 time=1.84 ms
64 bytes from 192.168.56.9: icmp_seq=3 ttl=64 time=3.44 ms
64 bytes from 192.168.56.9: icmp_seq=4 ttl=64 time=3.46 ms
```

2.2 Connectivity test for Local Machine 1 to Server 2: ✅ Successful ☐ Not Successful

```
queleroyubuntu3@workstation:~/Desktop$ ping 192.168.56.10
PING 192.168.56.10 (192.168.56.10) 56(84) bytes of data.
64 bytes from 192.168.56.10: icmp_seq=1 ttl=64 time=4.51 ms
64 bytes from 192.168.56.10: icmp_seq=2 ttl=64 time=2.05 ms
64 bytes from 192.168.56.10: icmp_seq=3 ttl=64 time=1.79 ms
```

2.3 Connectivity test for Server 1 to Server 2: ✅ Successful ☐ Not Successful

```
queleroyubuntu3@server1:~/Desktop$ ping 192.168.56.10
PING 192.168.56.10 (192.168.56.10) 56(84) bytes of data.
64 bytes from 192.168.56.10: icmp_seq=1 ttl=64 time=2.40 ms
64 bytes from 192.168.56.10: icmp_seq=2 ttl=64 time=1.91 ms
64 bytes from 192.168.56.10: icmp_seq=3 ttl=64 time=2.34 ms
64 bytes from 192.168.56.10: icmp_seq=4 ttl=64 time=2.04 ms
```

Task 4: Verify SSH connectivity on Server 1, Server 2, and Local Machine.

1. On the Local Machine, issue the following commands:

1.1 ssh username@ip_address_server1 for example, ssh jvtaylar@192.168.56.120

1.2 Enter the password for server 1 when prompted

```
queleroyubuntu3@workstation:~/Desktop$ ssh queleroyubuntu3@192.168.56.9
The authenticity of host '192.168.56.9 (192.168.56.9)' can't be established.
ED25519 key fingerprint is SHA256:pzvJhCv6fcwzMMivfjoIuBCc3wTBfmMzeMdxmfxR8Sw.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.9' (ED25519) to the list of known hosts.
queleroyubuntu3@192.168.56.9's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-136-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro
```

1.3 Verify that you are in server 1. The user should be in this format user@server1.
For example, jvtaylar@server1

```
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```

2. Logout of Server 1 by issuing the command control + D.

```
queleroyubuntu3@server1:~$
logout
Connection to 192.168.56.9 closed.
```

3. Do the same for Server 2.

```
queleroyubuntu3@workstation:~/Desktop$ ssh queleroyubuntu3@192.168.56.10
The authenticity of host '192.168.56.10 (192.168.56.10)' can't be established.
ED25519 key fingerprint is SHA256:BSUA5HXsyLauicMbBkSEkOMgHg0z3wnX2N1V2blcTjU.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.10' (ED25519) to the list of known hosts.
queleroyubuntu3@192.168.56.10's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-136-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

4. Edit the hosts of the Local Machine by issuing the command sudo nano /etc/hosts. Below all texts type the following:

4.1 IP_address server 1 (provide the ip address of server 1 followed by the hostname)

4.2 IP_address server 2 (provide the ip address of server 2 followed by the hostname)

```
queleroyubuntu3@workstation: ~/Desktop
  GNU nano 6.2                          /etc/hosts *
127.0.0.1       localhost
127.0.0.1       workstation
192.168.56.9    server1
192.168.56.10   server2
# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
```

4.3 Save the file and exit.

5. On the local machine, verify that you can do the SSH command but this time, use the hostname instead of typing the IP address of the servers. For example, try to do ssh jvtaylar@server1. Enter the password when prompted. Verify that you have entered Server 1. Do the same for Server 2.

Connecting to server1:

```
queleroyubuntu3@workstation:~/Desktop$ ssh queleroyubuntu3@server1
The authenticity of host 'server1 (192.168.56.9)' can't be established.
ED25519 key fingerprint is SHA256:pzvJhCv6fcwzMMivfjoIuBCc3wTBfmMzeMdxmfxR8Sw.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'server1' (ED25519) to the list of known hosts.
queleroyubuntu3@server1's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-136-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Thu Jul 30 08:50:39 2026 from 192.168.56.11
```

Connecting to server2:

```
queleroyubuntu3@workstation:~/Desktop$ ssh queleroyubuntu3@server2
The authenticity of host 'server2 (192.168.56.10)' can't be established.
ED25519 key fingerprint is SHA256:BSUA5HXsyLauicMbBkSEkOMgHg0z3wnX2N1V2blcTjU.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:4: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'server2' (ED25519) to the list of known hosts.
queleroyubuntu3@server2's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-136-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Jul 30 08:53:02 2026 from 192.168.56.11
```

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2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Reflections:

Answer the following:

1. How are we able to use hostname instead of IP address in SSH commands?

- By using the DNS of the computer to get the IP address, we are able to use the hostname instead of the IP address. By using this method, we can easily connect the computers instead of writing a long IP address everytime.

2. How secured is SSH?

- This is very secure as it has a method of encrypting the data which are sent through the computers. By doing this, it will protect the information from unauthorized users. This is also good for remote accessing of computers.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **N/A**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **N/A**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **N/A**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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